

Sungwoo Yang

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EDUCATION

DigiPen Institute of Technology | Daegu, South Korea & Redmond, WA

Relevant Courses: Computer Graphics, Advanced C/C++, Game Implementation Techniques, Linear Algebra.

TECHNICAL SKILLS

Programming Languages: C, C++, GLSL

Engines: Custom Engine Architecture, OpenGL, SDL2, WebGL

Interpersonal Skills: Git, WSL (Windows Subsystem for Linux), Visual Studio Interpersonal Skills: Solution-Oriented, Adaptable, Technical Leadership, Cross-departmental Communication

PROJECTS

Astar | Tech Lead

Sept/2025 -Present

- Developed a custom game engine using C++, OpenGL, and SDL2, successfully transitioning the project foundation from the Raylib library.
- Implemented core rendering features including procedural meshes, exponential fog, toon shading, and shadow mapping.
- Designed engine-level systems such as dynamic SVG-based map parsing and a custom light reflection mechanic.

Acorn | QA

Mar/2025 -Jun/2025

- Conducted gameplay testing for a student game project, focusing on player movement, level flow, collision behavior, and overall playability.
- Created detailed bug reports with reproduction steps, severity levels, expected results, and actual results to help the development team resolve issues efficiently.
- Performed regression testing after bug fixes to verify stability and ensure that new changes did not introduce additional gameplay or technical issues.

LEADERSHIP EXPERIENCE

DigiPen Institute of Technology | Tech Lead | Daegu, South Korea

- Led technical planning for a student game project by organizing programming tasks, setting implementation priorities, and supporting team members during development.
- Communicated with designers and artists to translate gameplay ideas into achievable technical systems and engine features.
- Helped identify technical issues, review implementation progress, and prioritize fixes before project milestones.